

BML Switch — API Guide for AppInSnap

AppInSnap API Guide — BML Integration Switch (Phase 0 Sandbox)

Audience: AppInSnap mobile engineers

Product: Bank Al Murqarmah Integration Switch (AppInSnap → Switch → Mock iMAL)

Version: Phase 0 / 0.1.0-SNAPSHOT

Sandbox Base URL: `http://46.224.146.158:18080`

Host / IP: 46.224.146.158

Port: 18080

Local Docker (optional): `http://localhost:8080`

Related	Link
Test data sheet	APPINSNAP-TEST-DATA.md
Postman how-to	APPINSNAP-POSTMAN-GUIDE.md
Collection	BML-Switch-Phase0.postman_collection.json
OpenAPI	../openapi/assanpay-switch-v1.yaml
Live Swagger	http://46.224.146.158:18080/swagger-ui.html
Smoke script	./scripts/smoke.sh

Mock OTP is always 1234 in this sandbox.

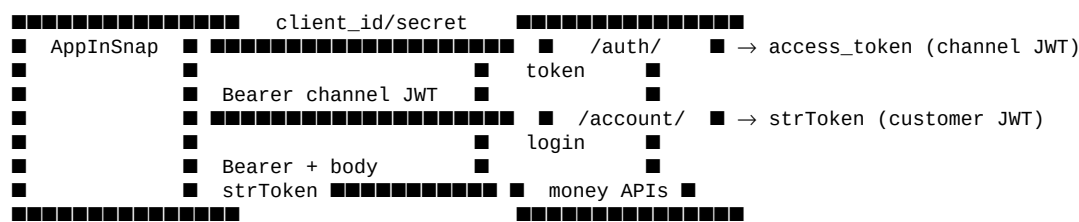
1. What this Switch is

The Switch is the **only backend** AppInSnap talks to for banking operations. It:

- Authenticates the **channel** (your app) with a short-lived JWT
- Authenticates the **customer** with `strToken` from login
- Serves **lookups** (banks, purpose codes, ...) so the app never hardcodes IMDs
- Proxies inquiry / IFT / IBFT to iMAL (mock in sandbox)
- Preserves AssanPay-compatible JSON shapes AppInSnap already expects

There is **no AppInSnap-facing frontend** — this is a REST API only.

2. Auth model (dual)



Token	Where it goes	Lifetime
Channel access_token	Header Authorization: Bearer ...	15 min (expires_in: 900)
Channel refresh_token	Body of /auth/refresh	7 days

Customer strToken	JSON body field strToken (required on customer APIs)	Same order as access (~15 min)
-------------------	---	--------------------------------

Optional alias: header X-Str-Token if body omits strToken. **Never** put strToken in Authorization.

2.1 Channel token

POST /api/v1/auth/token — **no Bearer required**

```
{
  "client_id": "appinsnap-sandbox",
  "client_secret": "change_me_sandbox_secret",
  "grant_type": "client_credentials"
}
```

Success (200):

```
{
  "access_token": "<jwt>",
  "token_type": "Bearer",
  "expires_in": 900,
  "refresh_token": "<opaque>",
  "refresh_expires_in": 604800,
  "correlationId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890"
}
```

Refresh: POST /api/v1/auth/refresh with { "refresh_token": "..." } — same response shape; old refresh is revoked.

Channel auth failure: HTTP **401** (problem JSON), e.g. detail: CHANNEL_TOKEN_EXPIRED.

2.2 Customer login → strToken

POST /api/v1/account/login — requires channel Bearer

```
{
  "UserName": "ali.khan",
  "CNIC": "4210112345678",
  "password": "Sandbox@123"
}
```

Success (200):

```
{
  "Response_Code": "00",
  "Response_Desc": "Success",
  "strToken": "<customer-jwt>",
  "UserName": "ali.khan",
  "correlationId": "..."
}
```

2.3 Standard headers on protected calls

```
Authorization: Bearer <access_token>
Content-Type: application/json
X-Correlation-Id: <uuid>           # optional; Switch mints if absent
Idempotency-Key: <uuid>           # required on IFT, IBFT, account/open
```

3. AssanPay response shape quirks (must parse correctly)

API	Nesting / casing	Notes
Balance	Capital Return	Return.Amount, Return.FromAccount, ...
IFT payment	Lowercase return	Nested object + top-level Response_Code

IBFT payment	Flat	stan, transactionID, fee at root — no return
IBFT title	Double nest	IBFTTitleFetchResponse.return...
Fee	Always "0.00" in mock	No separate fee API

Always check **HTTP status first**, then Response_Code when HTTP is 200. Auth failures are **401**, not soft 00.

4. Happy-path flows

4.1 Login → Balance → IFT

1. POST /auth/token → access_token
2. GET /lookups/banks (cache for later IBFT)
3. POST /account/login → strToken
4. POST /inquiry/balance (fromAccount=0345001234567)
5. POST /payment/ift with new Idempotency-Key → nested return.status=00

4.2 IBFT (with OTP ¹²³⁴)

1. Channel token + login (as above)
2. GET /lookups/banks → pick imd (e.g. 601004)
3. POST /inquiry/ibft-title with toIMD / toIBAN / toAccount
4. POST /auth/otp/send → otpReference
5. POST /auth/otp/verify with "otp": "1234" → otpTicket
6. POST /payment/ibft with otpTicket + Idempotency-Key → flat success + fee: "0.00"
7. Poll POST /inquiry/ibft-status until status=POSTED (sandbox is immediate)

4.3 Onboarding Path A (even CNIC — auto)

1. Channel token + lookups (occupations, purpose-of-account, provinces, ...)
2. Login (or register after open)
3. Upload / validate / liveliness / fingers (mock success for even CNIC)
4. POST /account/open + Idempotency-Key → accountNumber + IBAN
5. Register credentials if needed → login

4.4 Onboarding Path B (odd CNIC — manual)

1. Same KYC steps; odd last digit → open returns Response_Code 79
2. Sandbox only: POST /account/approve-mock with CNIC
3. Retry account/open

5. Test data (sandbox)

Full sheet: APPINSNAP-TEST-DATA.md

Item	Value
client_id / secret	appinsnap-sandbox / change_me_sandbox_secret
User / password	ali.khan / Sandbox@123
CNIC (even)	4210112345678
CNIC (odd / review)	4210112345679
From account	0345001234567 (150000 PKR)

IFT to	0345001234568 (50000 PKR)
Zero	0345009999999
Invalid	0345000000000
Mock OTP	1234
Sample IBFT IMD	601004 (Meezan)
Sample toIBAN	PK12MEZN0000001234567890

6. Field ownership

Owns	Examples
Mobile sends	strToken, amounts, accounts, CNIC, toIMD/toIBAN, OTP, PurposeOfPayment, Idempotency-Key, login password
Switch injects / overwrites	Bank IMD of own bank, agent account, channelId, MCC, merchant name, app id, delimiter, net id — do not trust client overrides
Switch generates if blank	stan / intSTAN, dates/times, correlationId

Amount format: string, 2 decimals, no commas → "100.00".

7. Endpoint catalog

Auth legend: **Public** | **Channel** (Bearer) | **Channel + strToken**

7.1 System

GET /api/v1/system/health — Public

Success:

```
{
  "status": "UP",
  "mockImal": true,
  "mockOtp": true,
  "db": "UP",
  "correlationId": "..."}

```

GET /api/v1/system/version — Channel

Returns appVersion, catalogVersion, mockImal, correlationId.

7.2 Auth

POST /api/v1/auth/token — Public

See §2.1. Errors: HTTP 401 invalid client.

POST /api/v1/auth/refresh — Public

```
{ "refresh_token": "<refresh_token>" }
```

POST /api/v1/auth/logout — Channel

```
{
  "refresh_token": "<optional>",
  "strToken": "<optional>"}
```

```

    "revokeCustomerSession": true
  }

```

Success: { "Response_Code": "00", "Response_Desc": "Success", "correlationId": "..."} }

POST /api/v1/auth/otp/send — Channel + strToken

```

{
  "strToken": "<customer-jwt>",
  "purpose": "IBFT",
  "mobile": "03001234567"
}

```

Success:

```

{
  "Response_Code": "00",
  "Response_Desc": "Success",
  "otpReference": "OTP-MOCK-A1B2C3D4",
  "expires_in": 120,
  "correlationId": "..."}

```

POST /api/v1/auth/otp/verify — Channel + strToken

```

{
  "strToken": "<customer-jwt>",
  "otpReference": "OTP-MOCK-A1B2C3D4",
  "otp": "1234"
}

```

Success:

```

{
  "Response_Code": "00",
  "Response_Desc": "Success",
  "otpTicket": "TICKET-...",
  "correlationId": "..."}

```

Common error (200): Response_Code 78 — wrong OTP / expired reference.

7.3 Lookups — Channel only

All return envelope:

```

{
  "correlationId": "...",
  "catalogVersion": 3,
  "asOf": "2026-09-28T...",
  "items": [ ... ],
  "Response_Code": "00",
  "Response_Desc": "Success"
}

```

Method	Path	Notes
GET	/api/v1/lookups/banks?active=true&supportsIbft=true	Required before IBFT UI — ≥16 banks
GET	/api/v1/lookups/banks/{imd}	Single bank; 404 if unknown
GET	/api/v1/lookups/purpose-of-payment	Alias: /purpose-codes
GET	/api/v1/lookups/purpose-of-account	
GET	/api/v1/lookups/occupations	
GET	/api/v1/lookups/response-codes	
GET	/api/v1/lookups/account-types	
GET	/api/v1/lookups/provinces	
GET	/api/v1/lookups/id-types	

GET	/api/v1/lookups/finger-indexes	
GET	/api/v1/lookups/onboarding-steps	
GET	/api/v1/lookups/branches	
GET	/api/v1/lookups/currencies	
GET	/api/v1/lookups/app-config	Public config map
GET	/api/v1/lookups/version	Catalog version

Sample bank item:

```
{
  "imd": "601004",
  "bankCode": "MEZN",
  "bankShortName": "Meezan",
  "bankName": "Meezan Bank Limited",
  "ibanBankCode": "MEZN",
  "supportsIbft": true,
  "isActive": true,
  "sortOrder": 40
}
```

Honor ETag / If-None-Match (304). **Do not hardcode bank lists in the app.**

7.4 Account / onboarding — Channel (+ strToken where noted)

POST /api/v1/account/login — Channel

See §2.2. Bad password → 14. Lockout → HTTP 401 + 75.

POST /api/v1/account/register — Channel

```
{
  "CNIC": "4210112345678",
  "AccountNo": "0345001234567",
  "MobileNo": "03001234567",
  "UserName": "ali.khan",
  "password": "Sandbox@123"
}
```

POST /api/v1/account/customer-detail — Channel + strToken

```
{ "strToken": "...", "CNIC": "4210112345678" }
```

Success nest: Response.CNIC, FULL_NAME, Email, ADDRESS, MOBILE_NUM, D_Birth.

POST /api/v1/account/account-list — Channel + strToken

```
{
  "strToken": "...",
  "Request": { "CNIC": "4210112345678" }
}
```

Success: Response.CASA_Account_List[] with Account_Number, Account_IBAN, Account_Title, ...

POST /api/v1/account/open — Channel + strToken + Idempotency-Key

```
{
  "strToken": "...",
  "CNIC": "4210112345678",
  "fullName": "ALI KHAN",
  "MobileNo": "03001234567",
  "occupationCode": "EMP",
  "purposeOfAccountCode": "SAV",
  "productCode": "ASAAN_DIGITAL"
}
```

Success:

```
{
  "Response_Code": "00",
}
```

```

    "Response_Desc": "Success",
    "accountNumber": "0345...",
    "IBAN": "PK00BMAL...",
    "accountTitle": "ALI KHAN",
    "currency": "PKR",
    "productCode": "ASAAN_DIGITAL",
    "status": "ACTIVE",
    "correlationId": "..."
  }

```

Odd CNIC without approve: 79 Manual review pending.

KYC mocks (Channel + strToken)

Path	Role
POST /account/upload-documents-unikrew	OCR + face confidence (even CNIC high)
POST /account/validate-document-unikrew	Doc authenticity mock
POST /account/verify-liveliness	Liveness mock
POST /account/verify-fingers	Biometric mock

POST /api/v1/account/approve-mock — Channel (sandbox only)

```

{ "CNIC": "4210112345679" }

```

POST /api/v1/account/change-password — Channel + strToken

```

{
  "strToken": "...",
  "currentPassword": "Sandbox@123",
  "newPassword": "Sandbox@456"
}

```

POST /api/v1/account/reset-password-mock — Channel (sandbox)

```

{ "CNIC": "4210112345678", "newPassword": "Sandbox@123" }

```

POST /api/v1/accounts/information — Channel + strToken

```

{ "strToken": "...", "fromAccount": "0345001234567" }

```

7.5 Payments — Channel + strToken + Idempotency-Key

POST /api/v1/payment/ift

```

{
  "strToken": "...",
  "fromAccount": "0345001234567",
  "toAccount": "0345001234568",
  "amount": "100.00",
  "PurposeOfPayment": "FAM",
  "Description": "smoke ift"
}

```

Success (nested return):

```

{
  "return": {
    "agentAccount": "00000000000",
    "amount": "100.00",
    "channelId": "MOBILE",
    "fromAccount": "0345001234567",
    "toAccount": "0345001234568",
    "stan": "123456",
    "status": "00",
    "statusDescription": "Success",
    "PurposeOfPayment": "FAM"
  },
  "Response_Code": "00",
  "Response_Desc": "Success",
  "correlationId": "..."
}

```

Common errors (HTTP 200): 51 NSF · 12/14 bad account · 61 daily limit · missing Idempotency-Key → 30.

POST /api/v1/payment/ibft

```
{
  "strToken": "...",
  "fromAccount": "0345001234567",
  "toIMD": "601004",
  "toIBAN": "PK12MEZN0000001234567890",
  "toAccount": "1234567890",
  "amount": "25.00",
  "PurposeOfPayment": "FAM",
  "Description": "smoke ibft",
  "otpTicket": "TICKET-..."
}
```

Success (flat):

```
{
  "Response_Code": "00",
  "Response_Desc": "Success",
  "stan": "654321",
  "transactionID": "TXN-0F753F3E-6B0",
  "fee": "0.00",
  "correlationId": "..."
}
```

Common errors: 78 OTP · 76 bad IMD · 51 NSF · 61 limit.

7.6 Inquiry — Channel + strToken

POST /api/v1/inquiry/balance

```
{ "strToken": "...", "fromAccount": "0345001234567" }
```

Success (capital Return):

```
{
  "Return": {
    "Amount": "150000.00",
    "BranchCode": "001",
    "FromAccount": "0345001234567",
    "CURRENCY_CODE": "PKR"
  },
  "Response_Code": "00",
  "Response_Desc": "Success",
  "correlationId": "..."
}
```

POST /api/v1/inquiry/ift-title

```
{ "strToken": "...", "Account": "0345001234568" }
```

Success: nested lowercase return with accountTitle, account, status, ...

POST /api/v1/inquiry/ibft-title

```
{
  "strToken": "...",
  "fromAccount": "0345001234567",
  "toIMD": "601004",
  "toIBAN": "PK12MEZN0000001234567890",
  "toAccount": "1234567890",
  "amount": "25.00"
}
```

Success (double nest):

```
{
  "IBFTTitleFetchResponse": {
    "return": {
      "accountTitle": "ALI KHAN",
      "toIBAN": "PK12MEZN0000001234567890",
      "toIMD": "601004",

```



```

        "toAccount": "1234567890",
        "fromAccount": "0345001234567",
        "amount": "25.00"
    }
},
"Response_Code": "00",
"Response_Desc": "Success",
"correlationId": "..."
}

```

POST /api/v1/inquiry/cnic

```
{ "strToken": "...", "CNIC": "4210112345678" }
```

Success: Res.GetAccountFromCNICItems.Item (object or array).

POST /api/v1/inquiry/ibft-status

```

{
  "strToken": "...",
  "transactionID": "TXN-...",
  "stan": "654321"
}

```

Success: Response_Code=00, status=POSTED, fee=0.00, amount, ...

POST /api/v1/inquiry/mini-statement

```
{ "strToken": "...", "fromAccount": "0345001234567", "limit": 10 }
```

Success: transactions[] with bookingDate, narration, direction, amount, balanceAfter, stan.

POST /api/v1/inquiry/statement

```

{
  "strToken": "...",
  "fromAccount": "0345001234567",
  "fromDate": "2026-09-01",
  "toDate": "2026-09-28"
}

```

Max window 90 days; else 61.

POST /api/v1/inquiry/limits

```
{ "strToken": "...", "fromAccount": "0345001234567" }
```

Success: dailyDebitLimit, dailyDebitUsed, dailyDebitRemaining, maxBalance, currency.

POST /api/v1/inquiry/receipt

```

{
  "strToken": "...",
  "stan": "654321",
  "transactionID": "TXN-..."
}

```

8. Error handling (quick)

Kind	HTTP	What to branch on
Channel / session / lockout	401	problem detail or Response_Code 91 / 75
Validation	400	Response_Code 30 + fieldErrors
Business (NSF, OTP, IMD, ...)	200	Response_Code (51, 78, 76, 14, 61, 79, ...)
Upstream / fault	5xx / 99	Retry with same Idempotency-Key for money APIs

Selected codes:

Code	Meaning
00	Success
12	Invalid / closed account
14	Not found / bad credentials
30	Validation
51	Insufficient funds
61	Limit exceeded
75	Lockout
76	Invalid / inactive IMD
77	KYC incomplete
78	OTP required or invalid
79	Manual review pending
91	Session expired / missing strToken

Full detail: error-handling.md.

9. Rate limits & lockout

Control	Behavior
Login lockout	5 failed passwords → HTTP 401 + 75; ~30 min lock
Idempotency	Required on IFT/IBFT/open; replay within ~24h returns cached body
Lookups cache	Honor Cache-Control / ETag (sandbox ~300s)

10. Bring-up & verification

Sandbox (shared with ApplnSnap):

```
curl -s http://46.224.146.158:18080/api/v1/system/health  
# expect: "status":"UP"
```

Base URL: `http://46.224.146.158:18080` · Swagger: `http://46.224.146.158:18080/swagger-ui.html`

Import Postman collection (BML-Switch-Phase0.postman_collection.json) — `baseUrl` is already set to the sandbox IP/port — then run folder **01** — **Happy path**.

Local Docker (optional for Switch team):

```
cp .env.example .env  
docker compose up --build -d  
curl -s http://localhost:8080/api/v1/system/health  
BASE=http://localhost:8080 ./scripts/smoke.sh
```

11. Support contacts

Questions on contract drift or sandbox access: Bank Al Murqamah Switch team (via your project liaison).

Include `correlationId` from the failing response in every ticket.

BML Switch — Postman Guide

AppInSnap — Postman Collection Guide

Audience: AppInSnap mobile / QA engineers

Collection file: BML-Switch-Phase0.postman_collection.json

Companion API doc: APPINSNAP-API-GUIDE.md

Test data: APPINSNAP-TEST-DATA.md

Mock OTP = 1234 (never 123456).

1. Prerequisites

1. Switch running locally:

```
docker compose up --build -d
curl -s http://46.224.146.158:18080/api/v1/system/health
```

Expect "status": "UP".

2. Postman Desktop or Postman web with ability to import a collection JSON.

2. Import the collection

1. Open Postman → **Import**.

2. Choose **File** → select

docs/handoff/BML-Switch-Phase0.postman_collection.json

(or drag-and-drop the file).

3. Confirm collection name: **BML Switch Phase 0 — AppInSnap**.

4. Open the collection → **Variables** tab. You should see:

Variable	Default	Notes
baseUrl	http://46.224.146.158:18080	Sandbox IP 46.224.146.158 port 18080
clientId	appinsnap-sandbox	
clientSecret	change_me_sandbox_secret	Sandbox only
accessToken	(empty)	Auto-set by 01 auth/token
refreshToken	(empty)	Auto-set by token
strToken	(empty)	Auto-set by 03 account/login
otpReference	(empty)	Auto-set by OTP send
otpTicket	(empty)	Auto-set by OTP verify
imd	601004	Meezan sample from seed
toIban	PK12MEZN0000001234567890	Sample beneficiary
toAccount	1234567890	Sample beneficiary
fromAccount	0345001234567	Funded seed
toAccountIft	0345001234568	Funded seed
stan	(empty)	Auto-set after IBFT
transactionId	(empty)	Auto-set after IBFT

No separate Postman Environment file is required for local sandbox — collection variables are enough. For a shared team Environment, copy the same keys into an Environment named BML Switch Sandbox and select it in the top-right dropdown.

3. Happy-path run order

Run requests **top to bottom** inside folder **01 — Happy path (in order)**:

#	Request	Saves variables
1	system/health	—
2	auth/token	accessToken, refreshToken
3	lookups/banks	optionally peek imd (already default 601004)
4	account/login	strToken
5	inquiry/balance	—
6	payment/ift	—
7	inquiry/ibft-title	—
8	auth/otp/send	otpReference
9	auth/otp/verify (1234)	otpTicket
10	payment/ibft	stan, transactionId
11	inquiry/ibft-status	—
12	inquiry/limits	—
13	inquiry/receipt	—

Tip: Use Postman's **Collection Runner** on folder 01 – Happy path (in order) with iterations = 1.

Other folders (02 – Lookups, 03 – Negative checks) are optional after the happy path works.

4. How variables are chained

auth/token
■ ■ Tests script → accessToken, refreshToken

account/login
■ ■ Authorization: Bearer {{accessToken}}
■ ■ Tests script → strToken

auth/otp/send
■ ■ body.strToken = {{strToken}}
■ ■ Tests script → otpReference

auth/otp/verify
■ ■ body.otp = "1234"
■ ■ body.otpReference = {{otpReference}}
■ ■ Tests script → otpTicket

payment/ibft
■ ■ body.otpTicket = {{otpTicket}}
■ ■ body.toIMD = {{imd}} (601004)
■ ■ Tests script → stan, transactionId

inquiry/ibft-status / receipt
■ ■ use {{stan}} / {{transactionId}}

If a Tests script did not run (failed request), variables stay empty — fix the failing step and re-run from that point (or from **auth/token** if the channel JWT expired).

5. OTP reminder

Step	Value
Verify body	"otp": "1234"
Wrong OTP / missing ticket on IBFT	Response_Code 78
Sandbox flag	MOCK_OTP=true

6. Idempotency in Postman

IFT and IBFT requests set header:

```
Idempotency-Key: {{$guid}}
```

Each click generates a new UUID (new payment attempt). To test **replay**, freeze a UUID in the header (replace {{\$guid}} with a fixed value) and send the same request twice — second response should match the first success body.

7. Troubleshooting

Symptom	Likely cause	Fix
HTTP 401, CHANNEL_TOKEN_*	Missing/expired Bearer	Re-run auth/token ; check Authorization header
HTTP 401, Response_Code 91	Missing/expired strToken	Re-run account/login ; ensure body includes strToken
HTTP 401, 75	Lockout after 5 bad passwords	Wait 30 min or use reset-password-mock (sandbox)
Response_Code 78 on IBFT	No / invalid otpTicket	Run OTP send → verify with 1234, then IBFT
Response_Code 51	Insufficient funds	Use funded 0345001234567, smaller amount; avoid zero account
Response_Code 76	Bad / inactive IMD	Use 601004 or pick from lookups/banks
Response_Code 14	Bad account / credentials	Check seed accounts & password Sandbox@123
Connection refused	Stack down	docker compose up --build -d
Empty accessToken after token	Tests script not executed	Open request → Tests tab; ensure 200 + JSON access_token

8. Automated alternative

Without Postman:

```
./scripts/smoke.sh
```

Same happy path (OTP 1234) against `http://localhost:8080`.

9. Sharing with the team

1. Share this folder: docs/handoff/ (at least README, API guide, Postman guide, test data, collection JSON).
2. Tell engineers: **import collection** → **run folder 01** – Happy path **in order** → **OTP is 1234**.
3. Point them to APPINSNAP-API-GUIDE.md for request/response contracts.

BML Switch — Sandbox Test Data

AppInSnap — Sandbox Test Data

Copy/paste sheet for Slack, email, or ticket comments.

Environment: shared sandbox · Base URL `http://46.224.146.158:18080` · IP `46.224.146.158` · Port `18080` · Mock OTP `1234`

Channel (AppInSnap client)

Field	Value
client_id	appinsnap-sandbox
client_secret	change_me_sandbox_secret
grant_type	client_credentials
Access token TTL	900 seconds (15 min)
Refresh token TTL	604800 seconds (7 days)

Customer login

Field	Value
Username	ali.khan
Password	Sandbox@123
CNIC (even → auto KYC)	4210112345678
Mobile	03001234567
Full name	ALI KHAN
Odd CNIC (manual review demo)	4210112345679

After login, use body field `strToken` on all customer APIs.

Accounts & balances (seed)

Role	Account	IBAN	Opening balance (PKR)
From (primary)	0345001234567	PK00BMAL0000000345001234567	150000.00
IFT to (same customer)	0345001234568	PK00BMAL0000000345001234568	50000.00
Zero / NSF tests	0345009999999	PK00BMAL0000000345009999999	0.00
Invalid (not found)	0345000000000	—	—

Currency: **PKR**. Fee in mock responses: always `0.00`.

Suggested test amounts

Scenario	Amount
----------	--------

Happy IFT	"100.00"
Happy IBFT	"25.00"
NSF (from zero acct)	"10.00" from 0345009999999
Large (still under daily limit)	"1000.00"

Daily debit limit (Asaan display): 200000.00 PKR. Max balance display: 1000000.00.

Mock OTP

Field	Value
Code	1234 (never 123456)
Flag	MOCK_OTP=true
Flow	POST /auth/otp/send → otpReference → POST /auth/otp/verify with otp": "1234" → otpTicket
IBFT without ticket	Response_Code 78

Sample beneficiary banks (IMD) — use Lookups, do not hardcode in app

IMD	Short	Bank	IBAN bank code
601004	Meezan	Meezan Bank Limited	MEZN
601001	HBL	Habib Bank Limited	HABP
601002	UBL	United Bank Limited	UNIL
601003	MCB	MCB Bank Limited	MUCB
627000	BML	Bank Al Murqamah (own)	BMAL

Postman / smoke sample beneficiary

Field	Value
toIMD	601004
toIBAN	PK12MEZN00000001234567890
toAccount	1234567890

Load full list: GET /api/v1/lookups/banks?active=true&supportsIbft=true (≥16 Pakistan banks).

Purpose of payment (examples)

Code	Description
FAM	Family Support
SAL	Salary
GDS	Goods Payment
EDU	Education
MED	Medical

Source of truth: GET /api/v1/lookups/purpose-of-payment.

Idempotency-Key

Rule	Detail
Required on	POST /payment/ift, POST /payment/ibft, POST /account/open
Format	UUID generated by mobile per logical attempt
Header	Idempotency-Key: <uuid>
Replay	Same key + same outcome within ~24h returns cached success body

Auth headers (reminder)

Authorization: Bearer <channel_access_token>
Content-Type: application/json
X-Correlation-Id: <optional-uuid>
Idempotency-Key: <uuid> # money / open only

Customer token goes in JSON body as strToken (not in Authorization).

Quick links

- API guide: APPINSNAP-API-GUIDE.md
- Postman: APPINSNAP-POSTMAN-GUIDE.md
- Collection: BML-Switch-Phase0.postman_collection.json